GSEO
GIANT ELECTRONIC OPTICAL

LENS CURVE FITTING

To enhance the yield rate and lenses quality, I cooperate with the team to design a software system that can compare the differences between design surface type coefficient and real lens surface type coefficient. The differences we find can be used for modifying the tooling to enhance our yield rate.

INTRODUCTION & METHOD

We use specific machine (e.g. UA3P) to attain all surface data and then fit the lens data into a curve to look for the asphere function.

We compare the curve with the design data designing by Zemax. If the coefficient is dramatic changed, we can adjust the plastic forming machine temperature, pressure, and cycle time to optimize our lens sets performance (e.g. MTF, lower color aberration, decrease lens shrink issues).

Fitting Process

Select Design file
Load design and tooling data

↓

File Form Recognition
Different form with different loading method

↓

Shift RT to Center
Ensure the placement

↓

Asphere Curve Fitting
Find the corresponding coefficient
1.Linear Polynomial fitting
2.Nonlinear Polynomial fitting

↓

Plot
Show RT/Error/Sag Figure

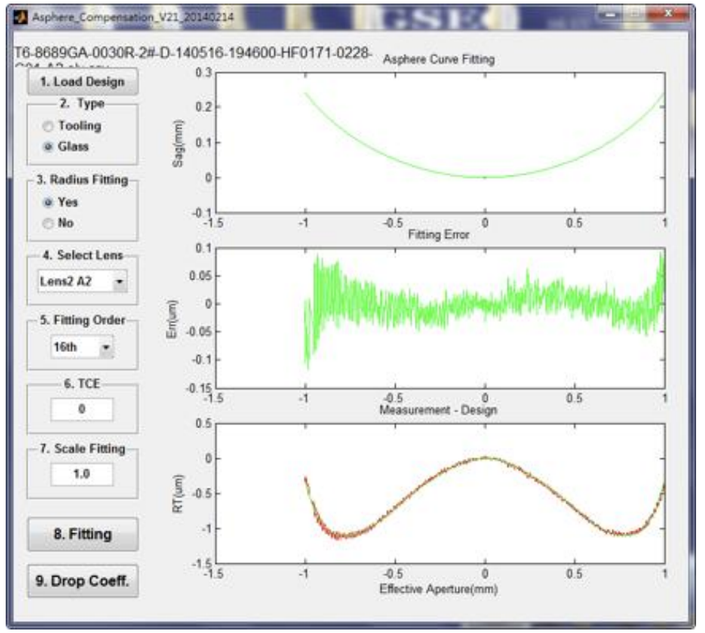
Asphere Function

$$Sag = \underbrace{\frac{cx^2}{1+\sqrt{1-(1+k)c^2x^2}}}_{\text{Nonlinear Term}} + \underbrace{C_4 x^4 + C_6 x^6 + C_8 x^8 + \cdots}_{\text{Linear Terms}}$$

Polyfit —> linear polynomial

Lsqcurvefit / Nlinfit —> nonlinear polynomial

The system allow users to choose specific lens side, fitting order to procure the fitting function.



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LIGHTHOUSE FUNDAMENTAL STRCUTURE

To realize immersive experiences for users, precise and accurate tracking system is needed. HTC Vive uses outside-in approaches to achieve the goal. The lighthouse is elaborate projector rendering infrared LED and laser sources for HMD and controllers' capture. Two lighthouses are for 360 degrees localization. If there is only one lighthouse, it will lose tracking when users turn their backs on a lighthouse.

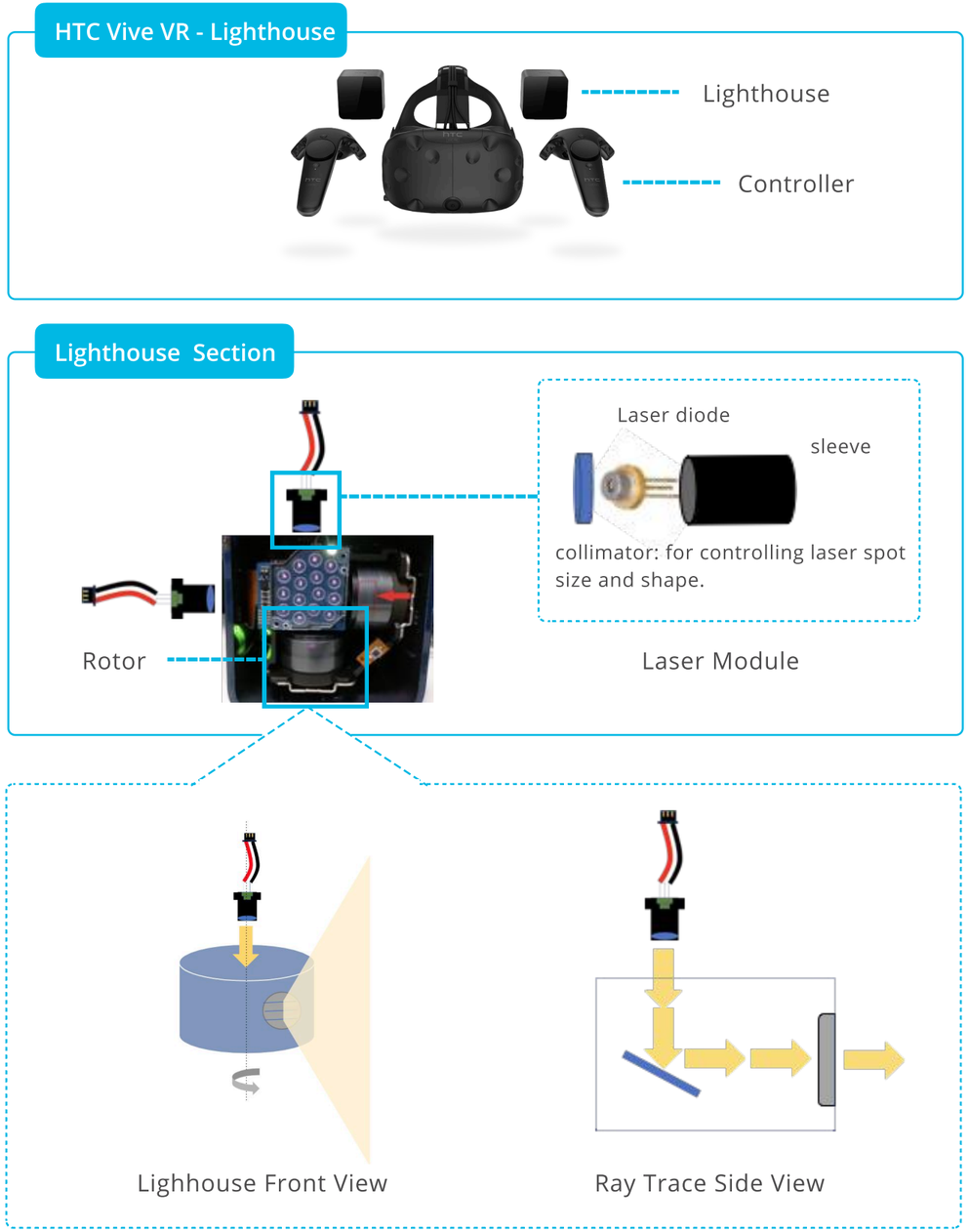
LIGHTHOUSE

The lighthouse is comprised of many optical components. Every component impacts other associated elements. The Laser diode pumps elliptical beam spot and pass through collimator. Laser beam spot change its ray trace by reflective mirror. After changing ray trace, the laser spot enter fan beam lens and then generate over 120 degree wide angle line beam for head mounted device to capture the signal. The two wide angle line beams perpendicularly and alternatively project onto the HMD. The HMD has a number of sensors to calculate the user's position.
The key and fundamental performance impact factors are (if assembly within acceptable tolerance):

A. The coaxiality between laser diode and collimator

B. The reflective mirror surface quality (scratch dot strip)

C. Fan beam lens surface quality and surface type

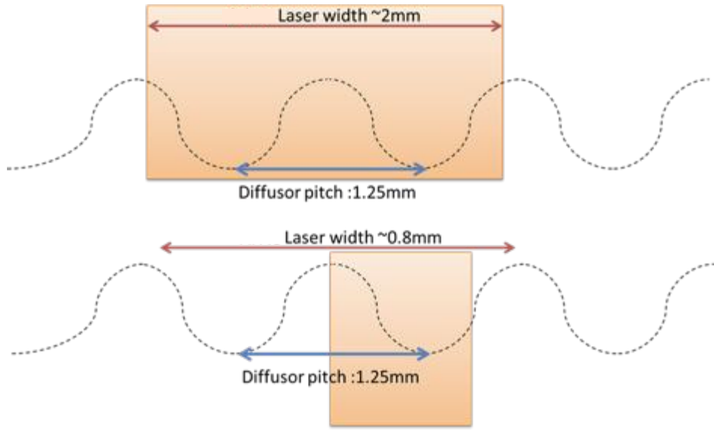


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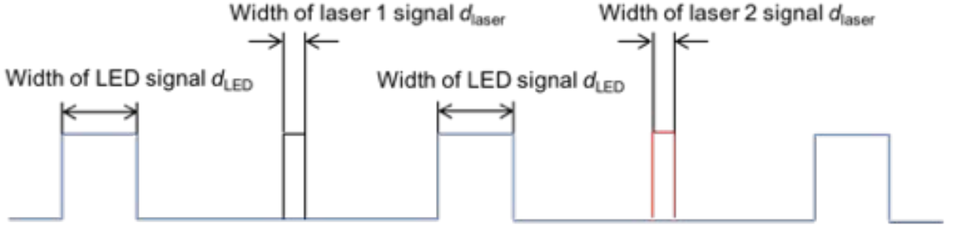
BEAM SHAPE, SIZE, AND POWER DISTRIBUTION

The spot size has pivotal relationship with the groove pitches. If line beam has high uniformity and less broken points due to interference phenomena, the whole tracking performance will be better (e.g. less jitter and loss tracking) and thus bringing about comfortable experience for users. The width of line beam influences laser signal time duration, so we have to regulate laser characters to a proper range.

Evaluate the relation between fan beam lens and laser spot

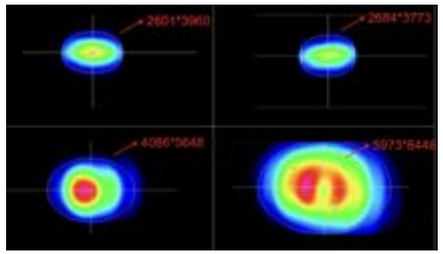


Laser and LED source in lighthouse signal behavior

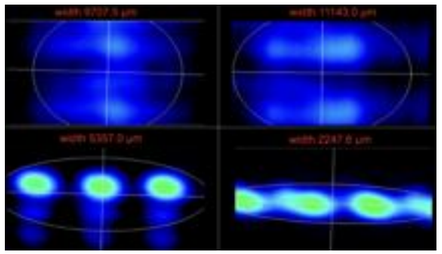


The line beam width influences the hitting time duration what the HMD sensors recognized

Data analysis and Evaluation

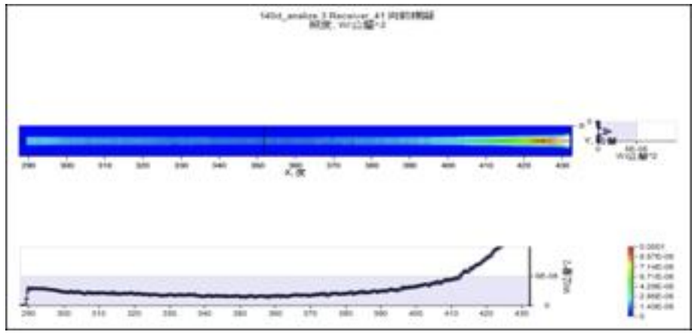


Laser spot size and power distribution



Line beam power distibution and shape test result

We attempt to adjust laser spot size and groove pitches of fan beam lens to assure high uniform line beam.

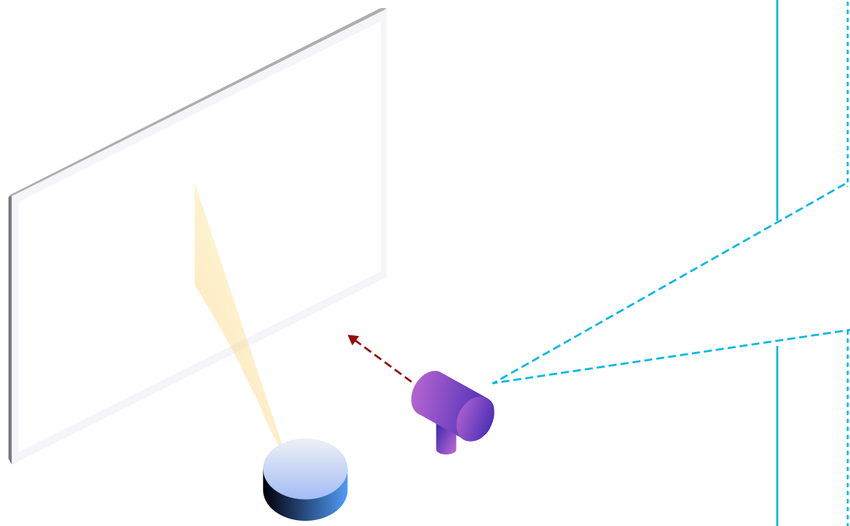


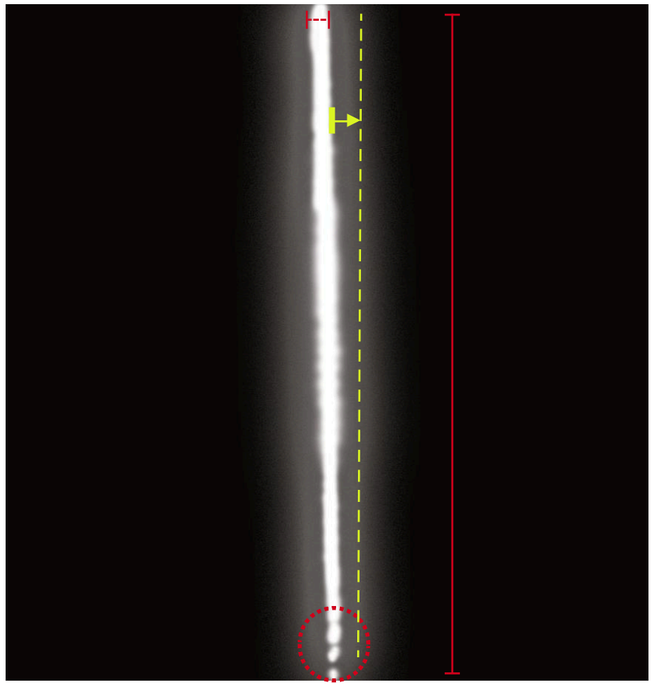
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LINE BEAM EVALUATION TOOL DESIGN

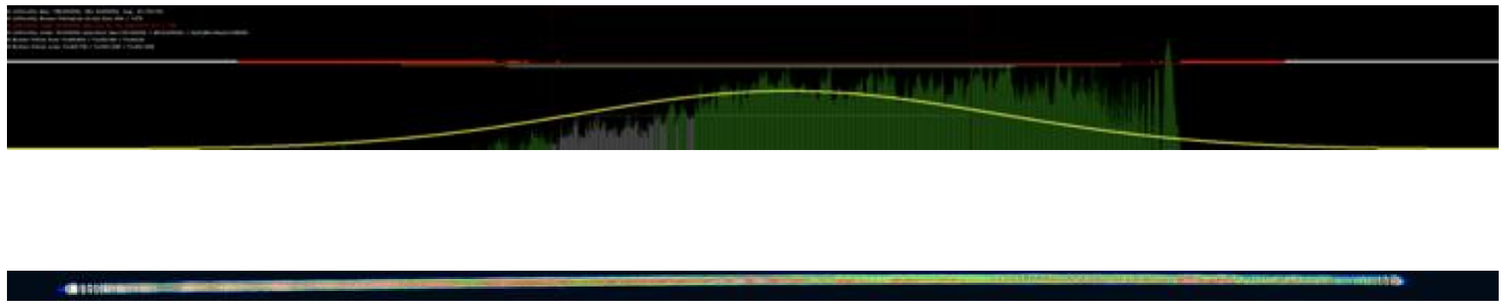
To find out better laser beam shape and power distribution, I design the tool to evaluate line beam width, length, linearity and power distribution. The tool help to collect key data and do basic analytics to correlate the components data with system data. (e.g. CPK analysis, normal distribution analysis)

System Design





Line beam power distribution analysis



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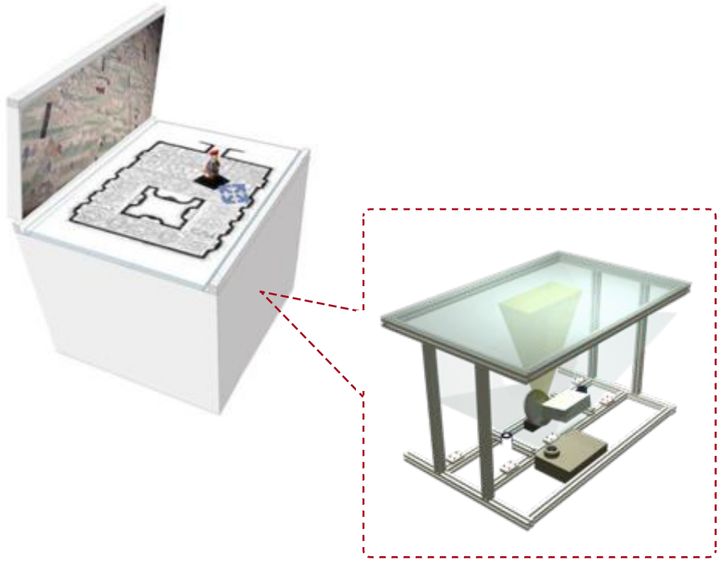
I-AM-TOP AUGMENTED REALITY PLATFORM

In this project, I-m-Top, I'm responsible for establishing a high quality rear projection system which is perpendicular to multi-point touch screen. The polarization stacks at transceiver and receiver (glasses) can significantly reduce color shift problem and crosstalk. It offers better visual experiences on 3D information.

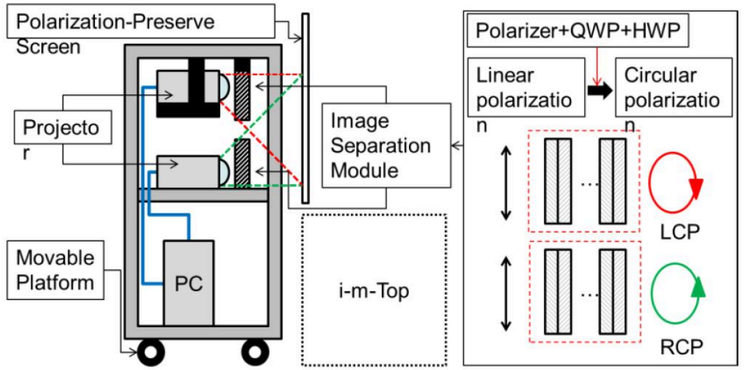
The augmented reality system combines :

1. 3D browsing system
2. Develop a new chip set for 60-GHZ transceiver with the ultimate integration goals of the system-on-chip (SoC) and the system-in-package (Sip)
3. Cloud-based intelligent for personalized health-care

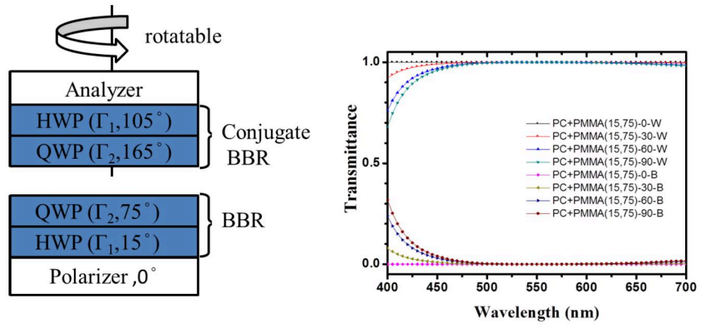
Overall System



System Structure



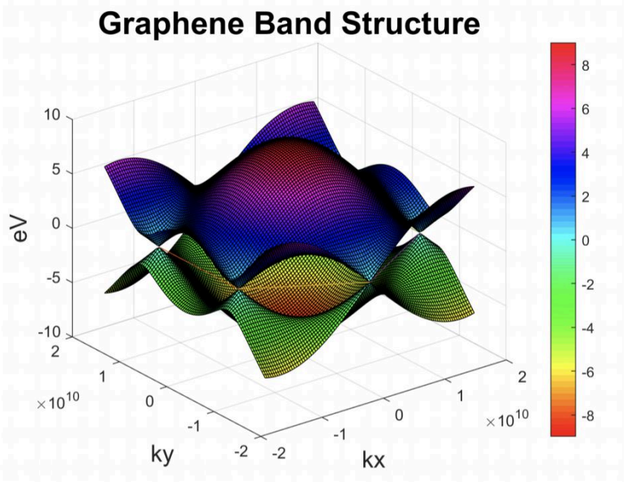
Stacked Retarders

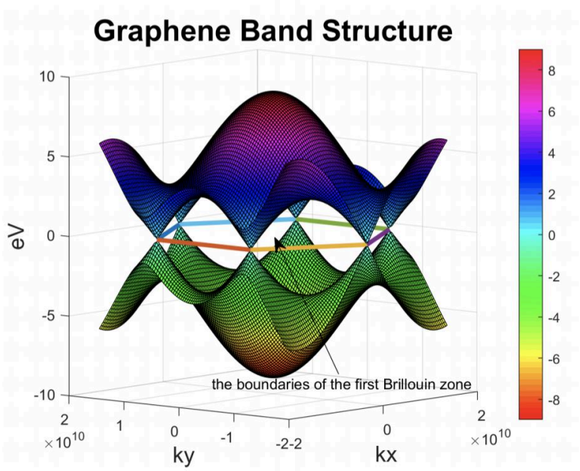


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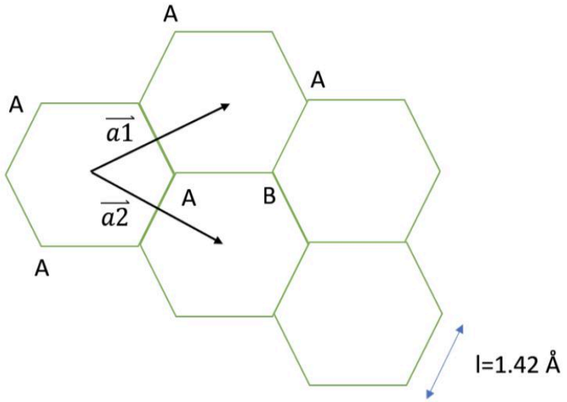
2D MATERIAL: Structure Visualization

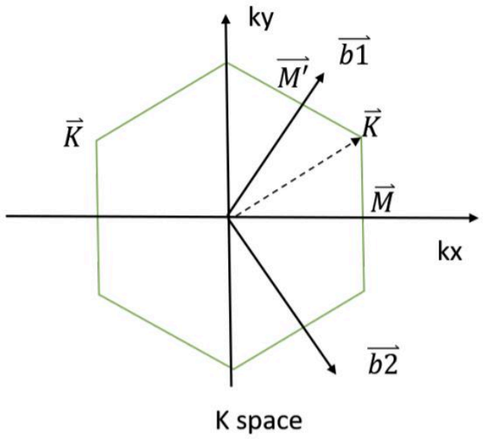
Visualized bandstructure of graphene (Energy E versus wave-vector k) by using the tight-binding method





The Brillouin zone of graphene is a hexagon. The special points are G, which is the zone center, K, which is the corners of the hexagon, and M is the midpoint between two adjacent K. State the vectors G, K, M.





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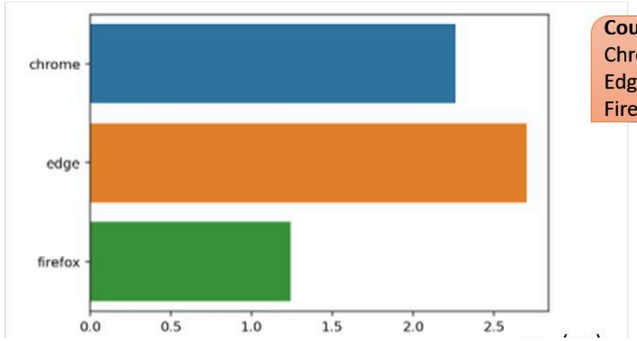
Web-usage applications and impacts on client resource utilization

Focused on web usage to acquire what browsers users prefer to use and analyzed the scenario by collecting data.

Calculated and visualized the parent and sub categories to summarize users' behavior and preferences. Correlated frame per second data and web usage data to learn how much data would be transferred per day. Analyzed what activities when users opened in a specific browser would bring the majority of data transmission. Bridged phenomenon of frame rate and CPU performance on different processes. Clarified the mapping between data rate and frame rate to capture useful information on the CPU part.

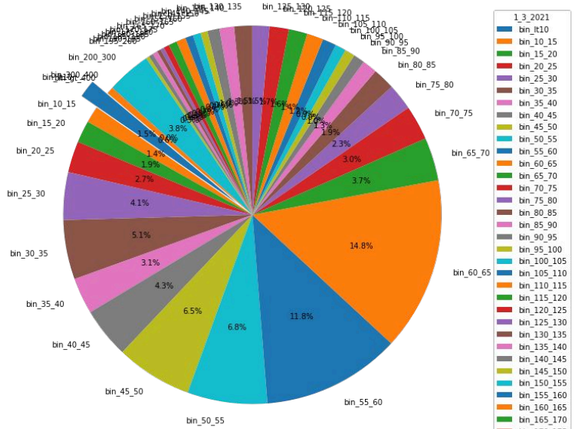
Mainly used Python Liberary Pandas, Numpy, Matplotlib, Seaborn to process data.

Browsers usage by frame data

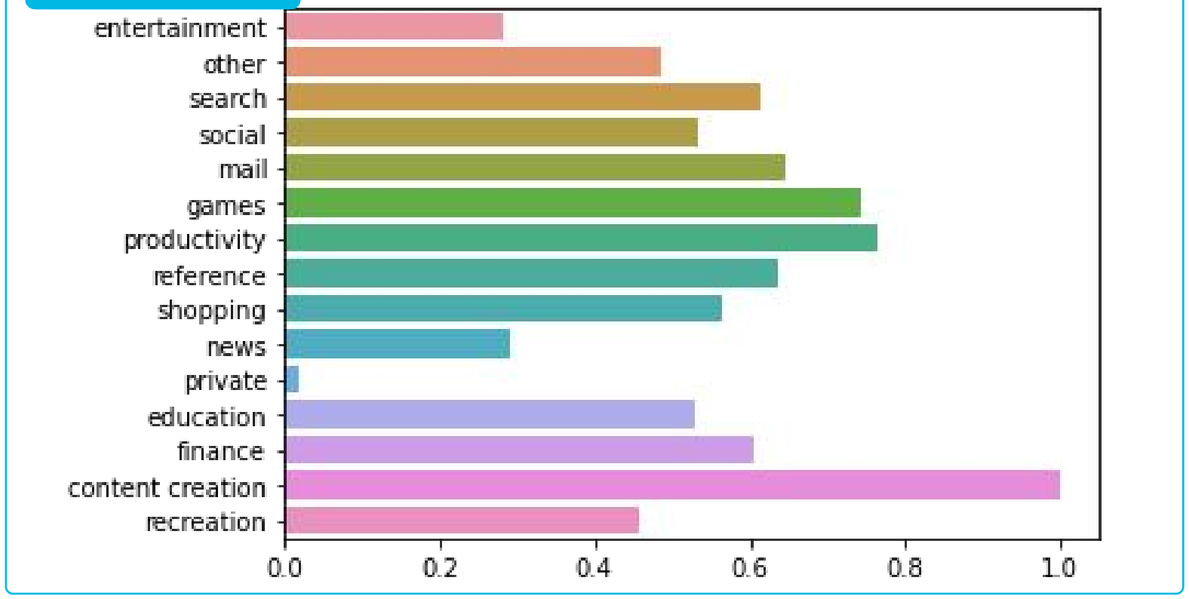


Count:
Chrome: 12221
Edge: 1018
Firefox: 395

~90% !



Browsers Purposes



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CNN Model

CNN Application on NBA Basketball Games

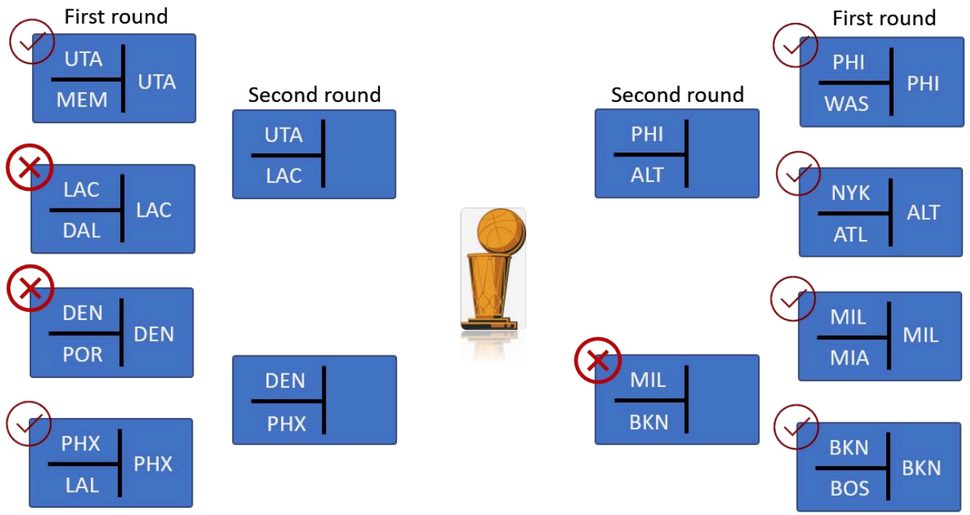
This experiment uses an unusual way to evaluate the win probability on NBA playoffs by players' performance. The playoffs are the battlefield of stars . Key players may represent the results.

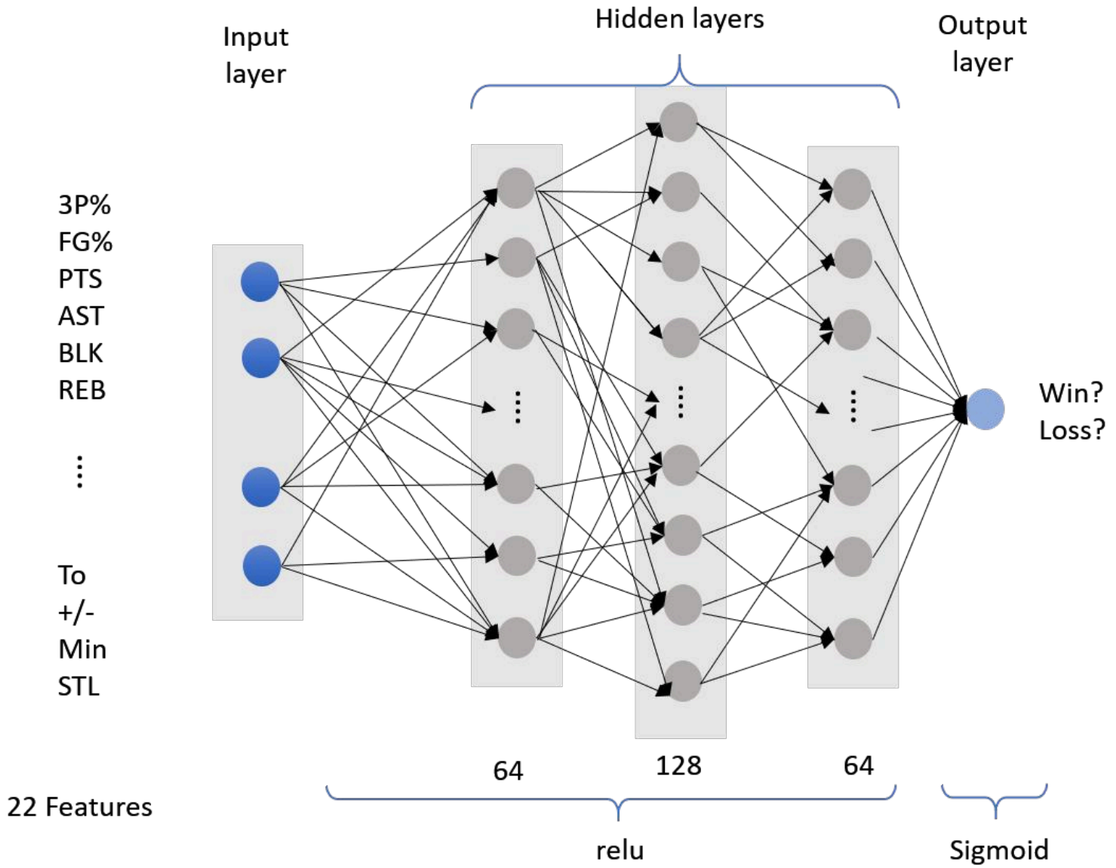
Data crawler: Beautiful Soup. The number of data: over 50 thousands

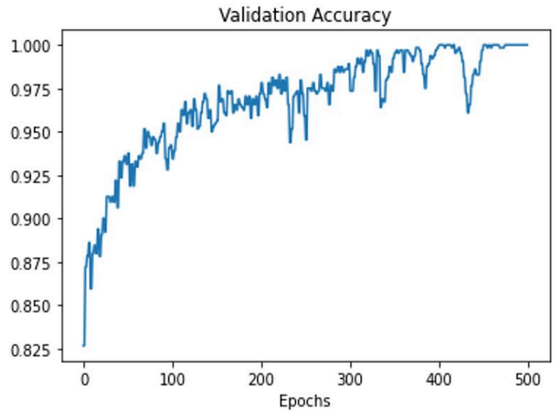
Collect and preprocess ten years(2011 2021) 16 Playoffs teams' rosters' traditional data.

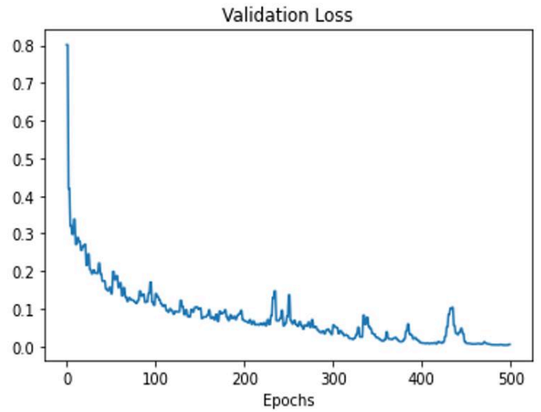
Choose top impact players. Shuffle before training.

Construct Deep Neutral Networks model to predict win or loss by parts of players in a team









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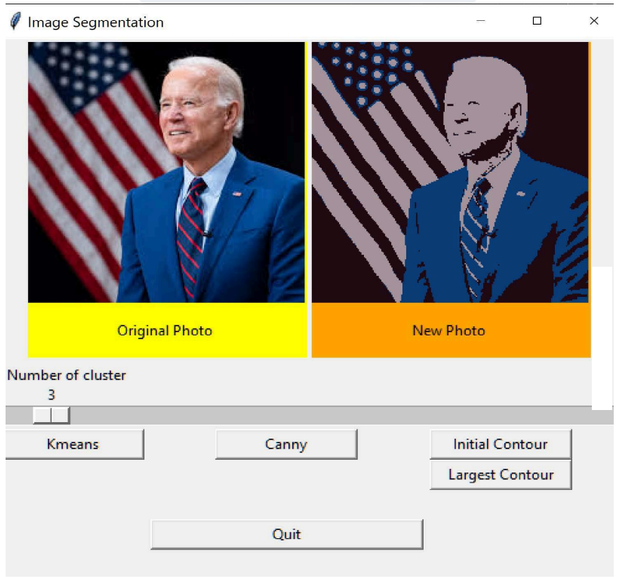
Image Processing

K-Means Clustering: Image Contour on human's head and body

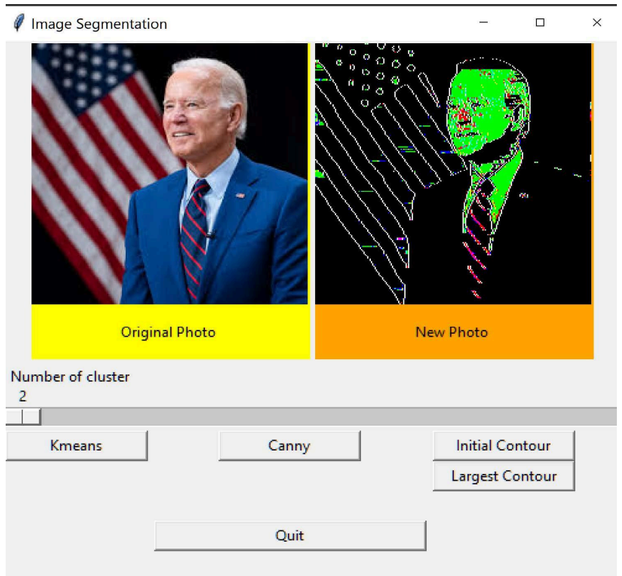
Use K-mean clustering pictures to look for obvious targeted areas with tkinter liberary for generating user interface.

The task of grouping pixel into two or more groups based on the value of the pixel. Here we can set up iteration times and the number of data. Default is 10. The canny function is to find the intensity gradient of the image. Filter image with a Sobel kernel in both horizontal and vertical direction to get first derivative because gradient direction is always perpendicular to edges.

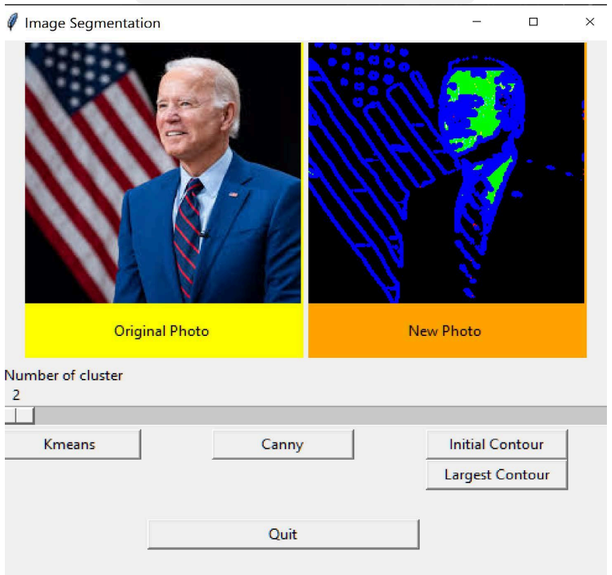
K-means



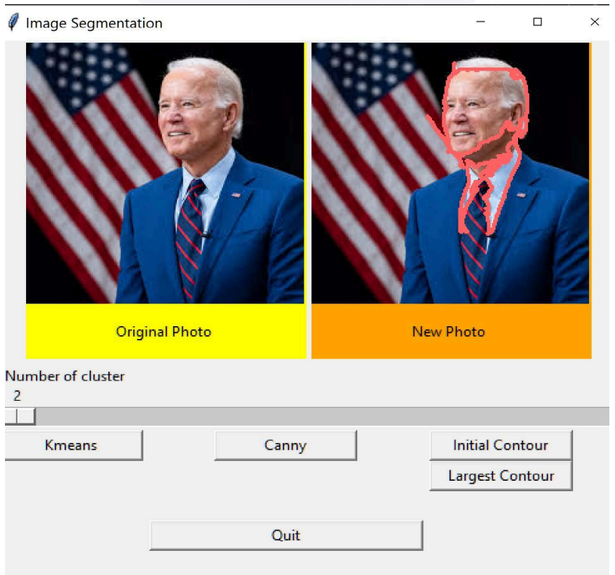
Danny Edge Detection



Contour 1



Contour 2



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Full-Stack

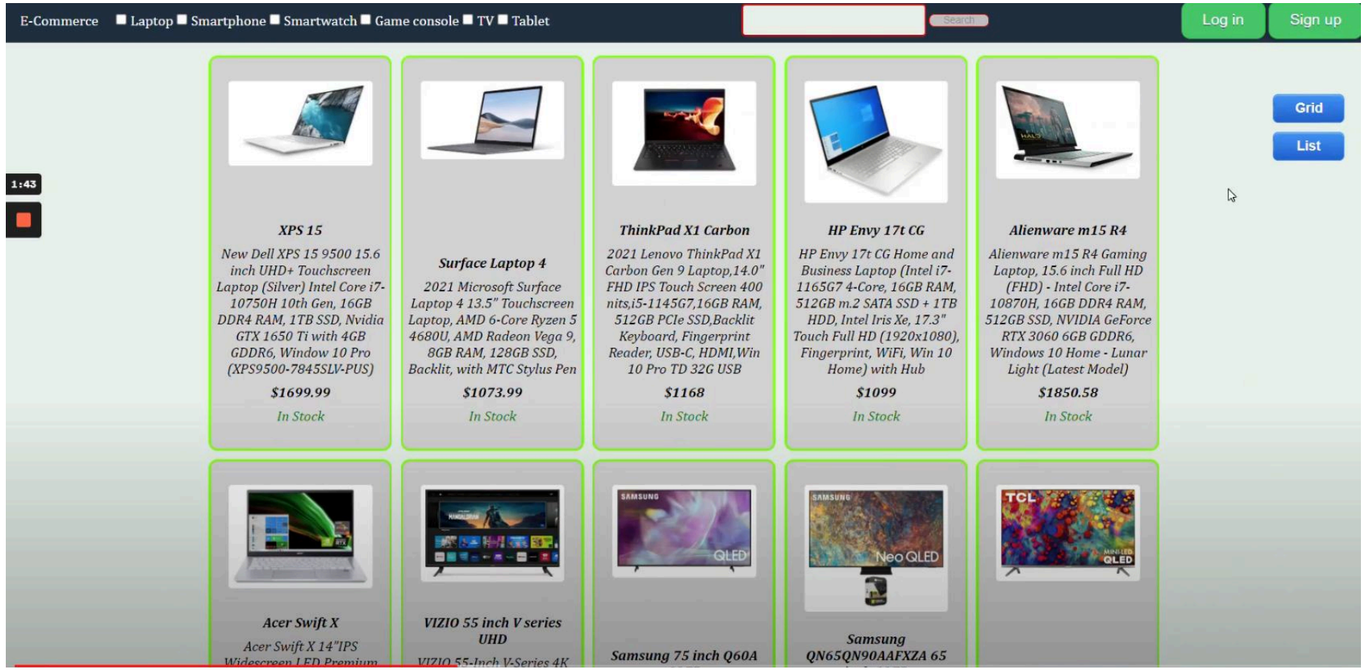
E-Worlds, a web application includes front-end and back-end design

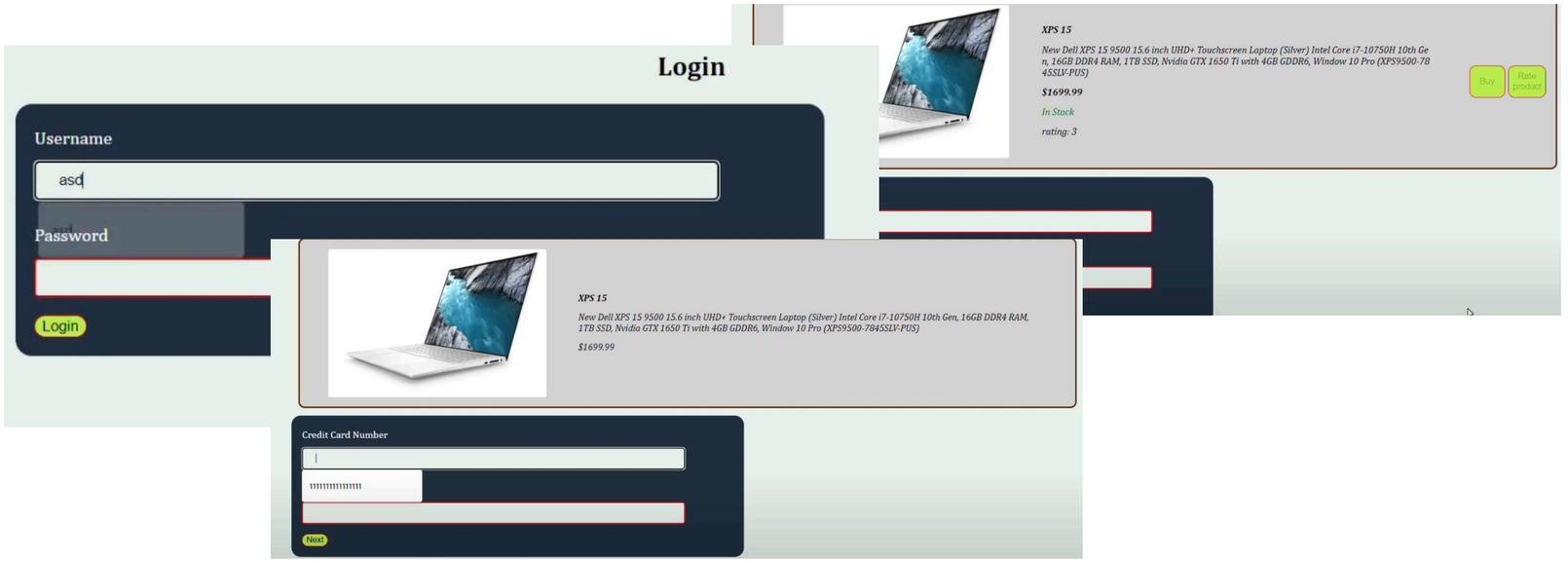
Designed a responsive MVC shopping website including sign-in, rating, transactions and categories filters implemented by Express, Node.js and SQlite.

Created and built efficient SQL queries and schema to filter products for accelerating searching speed and listing correlated items.

```
        "product_id"    INTEGER,
        "name"  TEXT,
        "category"      BLOB,
        "price" FLOAT,
        "availability"  TEXT,
        "capacity"      TEXT,
        "descriptions"  TEXT,
        PRIMARY KEY("product_id" AUTOINCREMENT)

> .schema reviews
CREATE TABLE IF NOT EXISTS "reviews" (
        "product_id"    INTEGER,
        "username"      INTEGER,
        "date"  DATETIME DEFAULT (datetime('now', 'localtime')),
        "comment"       TEXT,
        "rating"        FLOAT,
        FOREIGN KEY("product_id") REFERENCES "products",
        FOREIGN KEY("username") REFERENCES "users"
);
sqlite> .schema transactions
CREATE TABLE IF NOT EXISTS "transactions" (
        "confirmation_number"   INTEGER,
        "username"      INTEGER,
        "product_id"    INTEGER,
        "date"  DATETIME DEFAULT (datetime('now', 'localtime')),
        FOREIGN KEY("username") REFERENCES "users",
        FOREIGN KEY("product_id") REFERENCES "products",
        PRIMARY KEY("confirmation_number" AUTOINCREMENT)
);
sqlite> .schema users
CREATE TABLE IF NOT EXISTS "users" (
        "email" TEXT,
        "username"      TEXT,
        "password"      TEXT,
        PRIMARY KEY("username")
```





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Intro: https://drive.google.com/file/d/1yWB3UCxkf-ibxkNDSDmdSBLBndKs9pth/view?usp=share_link

Youtube QA System

A retrieval-augmented question answering (RAG) system designed for YouTube videos

It extracts **transcripts** from videos, encodes them into **embeddings**, and retrieves relevant segments based on user queries. The system supports **contextual question answering**, **batch processing** of multiple queries, and returns answers with **timestamp** references, allowing users to trace responses back to specific video segments. It demonstrates practical integration of **LLMs[Google genmini]**, embedding-based search, and lightweight system design for real-world information retrieval tasks.

```
❓ QUESTION 1:
What is the main topic of this video?
--------------------------------------------------------------------------------
💬 ANSWER:
The main topic of this video is **Zion National Park in Utah**. The video mentions its jaw-dropping views, spectacular hikes, and top sights and attractions. [00:00:
00.000 - 00:00:55.760](https://youtu.be/WkILH17TYm4?t=0)

⏱️RELEVANT TIMESTAMPS:
  • 00:00:00.000

Note: Timestamps in blue are clickable links to jump to that point in the YouTube video.

🔗 PLAIN URLS FOR MACOS USERS:
  • 00:00:00.000: https://www.youtube.com/watch?v=WkILH17TYm4&t=0

Note: For macOS users, you can copy and paste these URLs into your browser.
================================================================================

❓ QUESTION 2:
What are the key points discussed in the video?
--------------------------------------------------------------------------------
💬 ANSWER:
The key points discussed in the video are:

* **Zion National Park in Utah has some of the most jaw-dropping views and spectacular hikes in America.** [00:00:00.000 - 00:00:12.320](https://youtu.be/WkILH17TYm4
?t=0)
* **The park features massive sandstone cliffs, narrow slot canyons, and a river below.** [00:00:14.160 - 00:00:19.039](https://youtu.be/WkILH17TYm4?t=14)
* **There's something to do for everyone in the park, whether you're an adventure seeker or a casual hiker.** [00:00:20.480 - 00:00:26.080](https://youtu.be/WkILH17T
Ym4?t=20)
* **The video will cover some of the top sights and attractions in Zion National Park, including scenic drives, picturesque overlooks, thrilling hiking trails, and c
ascading waterfalls.** [00:00:27.279 - 00:00:40.000](https://youtu.be/WkILH17TYm4?t=27)
* **One of the most famous and frightening trails in Zion National Park is the hike to Angel's Landing.** [00:00:46.000 - 00:00:55.760](https://youtu.be/WkILH17TYm4?
t=46)

⏱️RELEVANT TIMESTAMPS:
  • 00:00:00.000
  • 00:00:14.160
  • 00:00:20.480
  • 00:00:27.279
  • 00:00:46.000

Note: Timestamps in blue are clickable links to jump to that point in the YouTube video.
```

Questions Answers

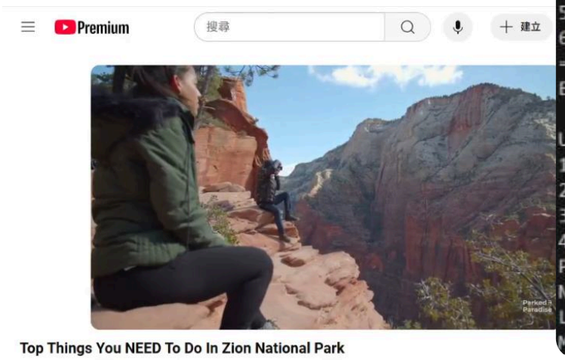
Timestamp reference

```
================================================
🎥 YOUTUBE TRANSCRIPT QA SYSTEM
================================================
1. Load YouTube video
2. Process questions (with context)
3. Display QA history
4. Clear cache
5. Use default questions
6. Exit
================================================================================
Enter your choice (1-6): 5

Using default questions:
1. What is the main topic of this video?
2. What are the key points discussed in the video?
3. What examples or demonstrations are provided?
4. What conclusions or recommendations are made?
Processing questions in batch...
Note: Each question will have access to the answers of previous questions
Loading embedding model (this may take a few seconds)...----------| 25.0% Complete
Model loaded successfully!
```

Options

Choose a YT video




```
================================================
🎥 YOUTUBE TRANSCRIPT QA SYSTEM
================================================
1. Load YouTube video
2. Process questions (with context)
3. Display QA history
4. Clear cache
5. Use default questions
6. Exit
================================================================================
Enter your choice (1-6): 1

📁 CACHED VIDEOS:
1. Top Things You NEED To Do In Zion National Park
2. Load a new video

Select a video (1-2) or enter a new YouTube URL:
```

Cached data

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Youtube QA System

High-level architecture of the Youtube QA system

Key blocks

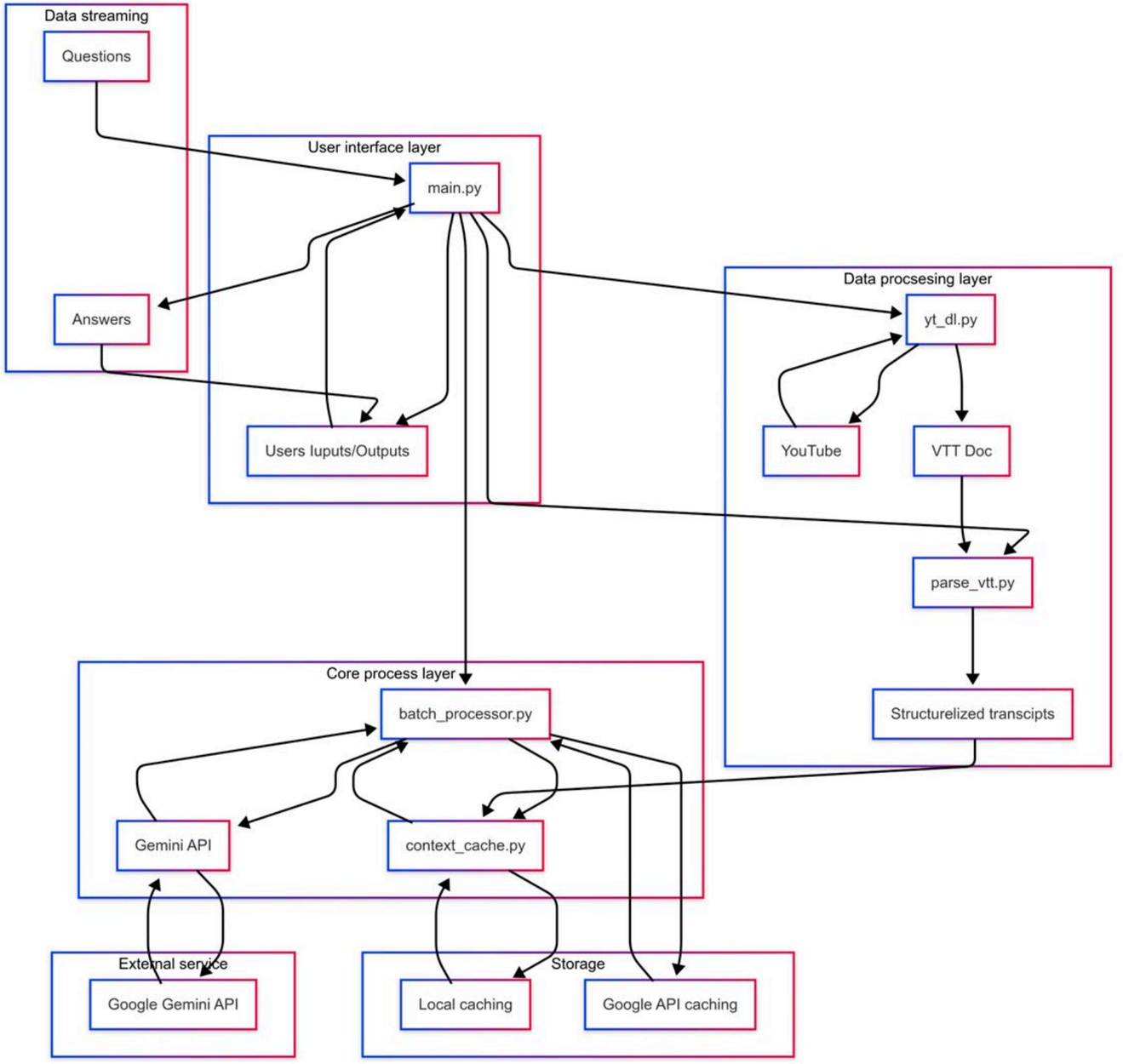
Data processing layer

User interface layer

Data processing layer

Core process layer

Storage



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Github: https://github.com/Hoogashaga/Batch-Prediction